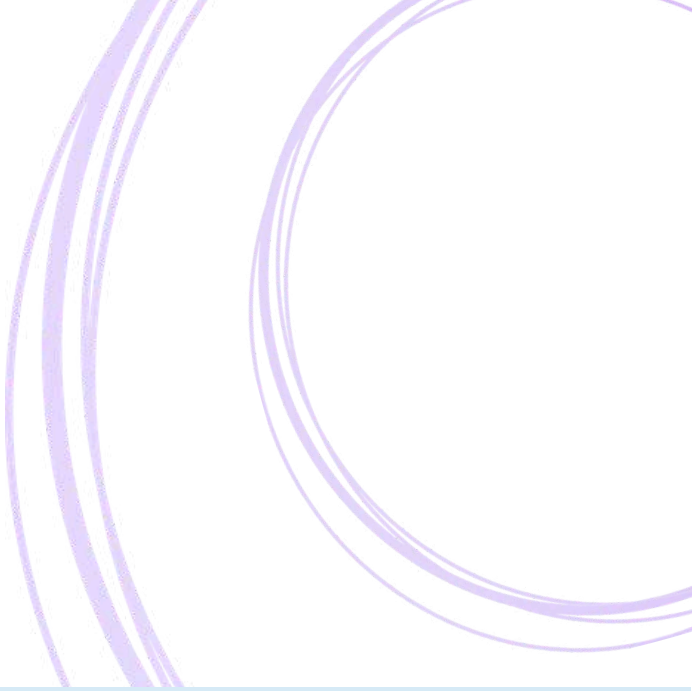
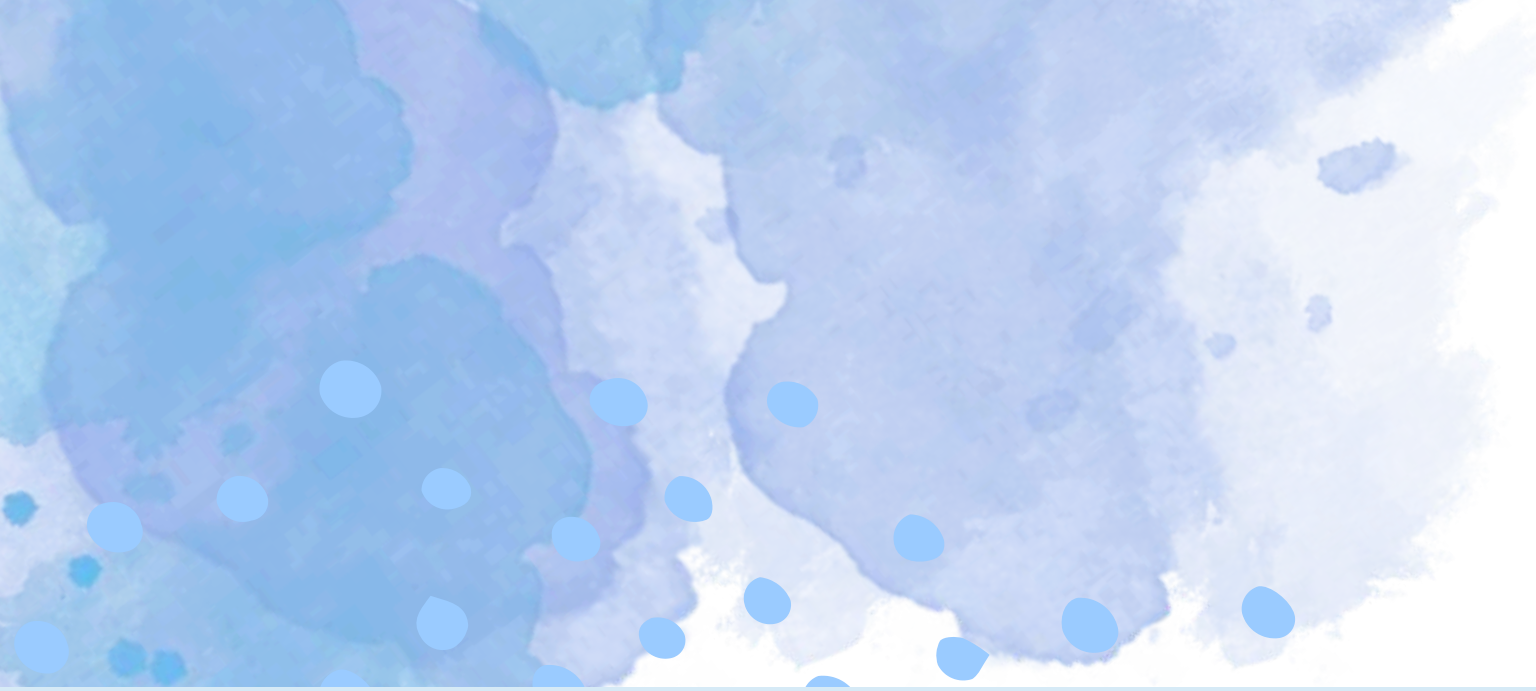




嘉義大飯店 訂房管理系統

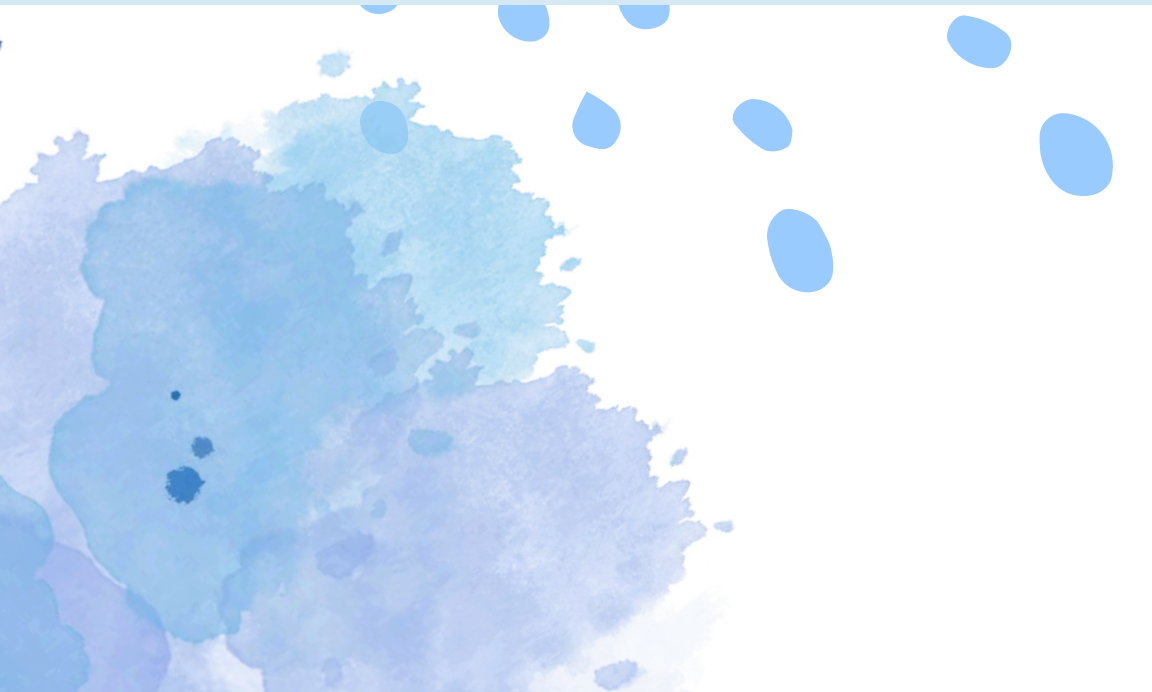
Team 10

1122959 邱采嫻

1122963 李月馨

1122967 林冠穎

1122968 歐家怡



系統架構設計



MVC 架構



- **View (視圖層):**
 - **Jinja2 (負責 HTML 結構渲染)**
 - **Bootstrap 5 (負責 UI 樣式與排版)**
- **Controller (控制層):**
 - **Flask (路由與邏輯控制)**
 - **jQuery (前端事件處理)**
- **Model (模型層):**
 - **PyODBC (資料庫連線)**
 - **SQL Server**

功能介紹



功能介紹

● 註冊 / 登入 / 登出

● 空房查詢

● 編輯 / 取消訂單

● 撰寫評論

● 管理者後台房間管理

● 管理者後台分析



SQL查詢語法



登入\註冊

使用者名稱、帳號及密碼不得為空

SQL

```
IF NOT EXISTS (SELECT * FROM sys.tables WHERE name = 'Users')  
BEGIN
```

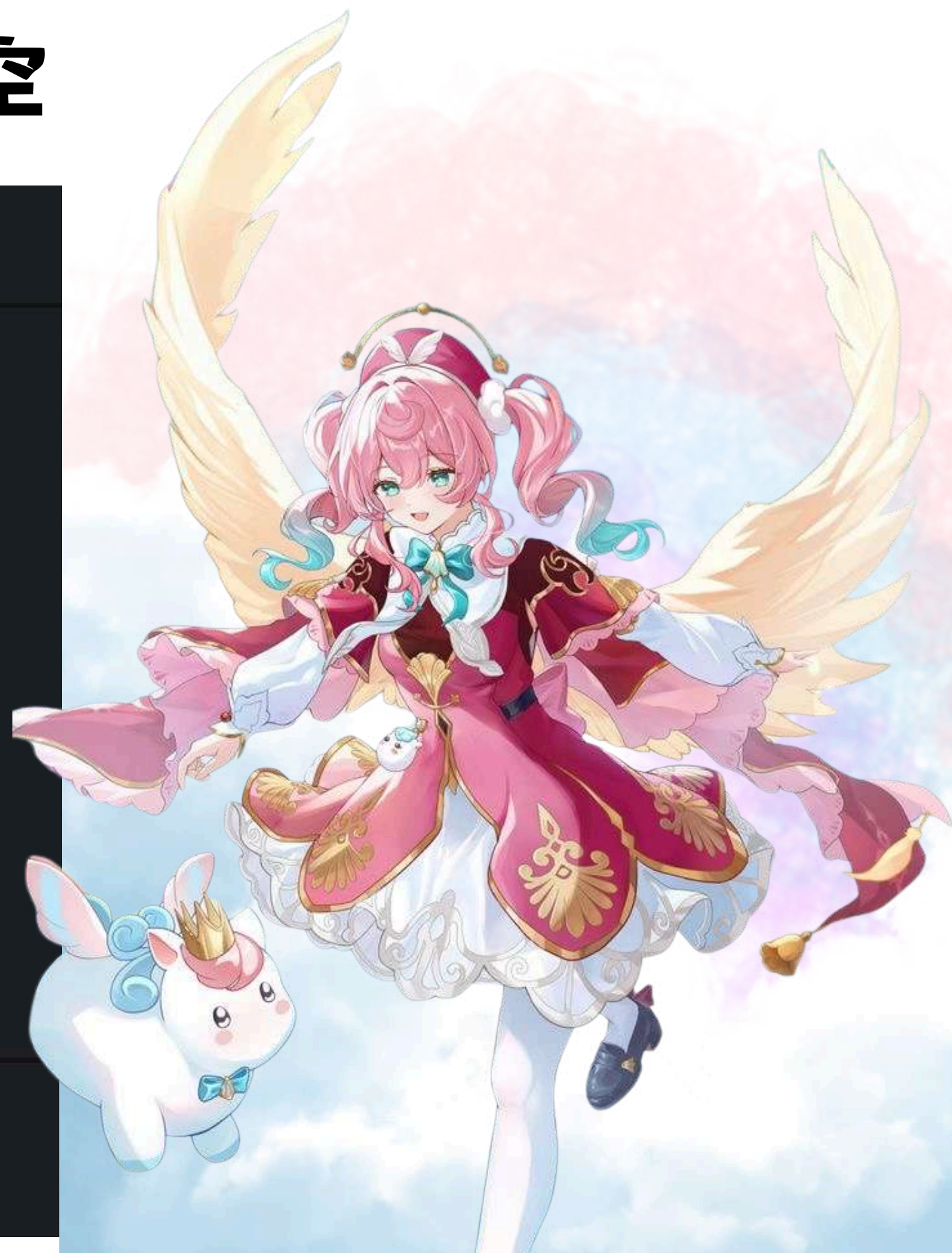
```
    CREATE TABLE Users (  
        user_id INT IDENTITY(1,1) PRIMARY KEY,  
        user_name NVARCHAR(100) NOT NULL,  
        account NVARCHAR(100) NOT NULL UNIQUE,  
        password NVARCHAR(100) NOT NULL  
    );
```

END

GO

SQL

```
INSERT INTO Users (user_name, account, password)  
VALUES (@user_name, @account, @password);
```

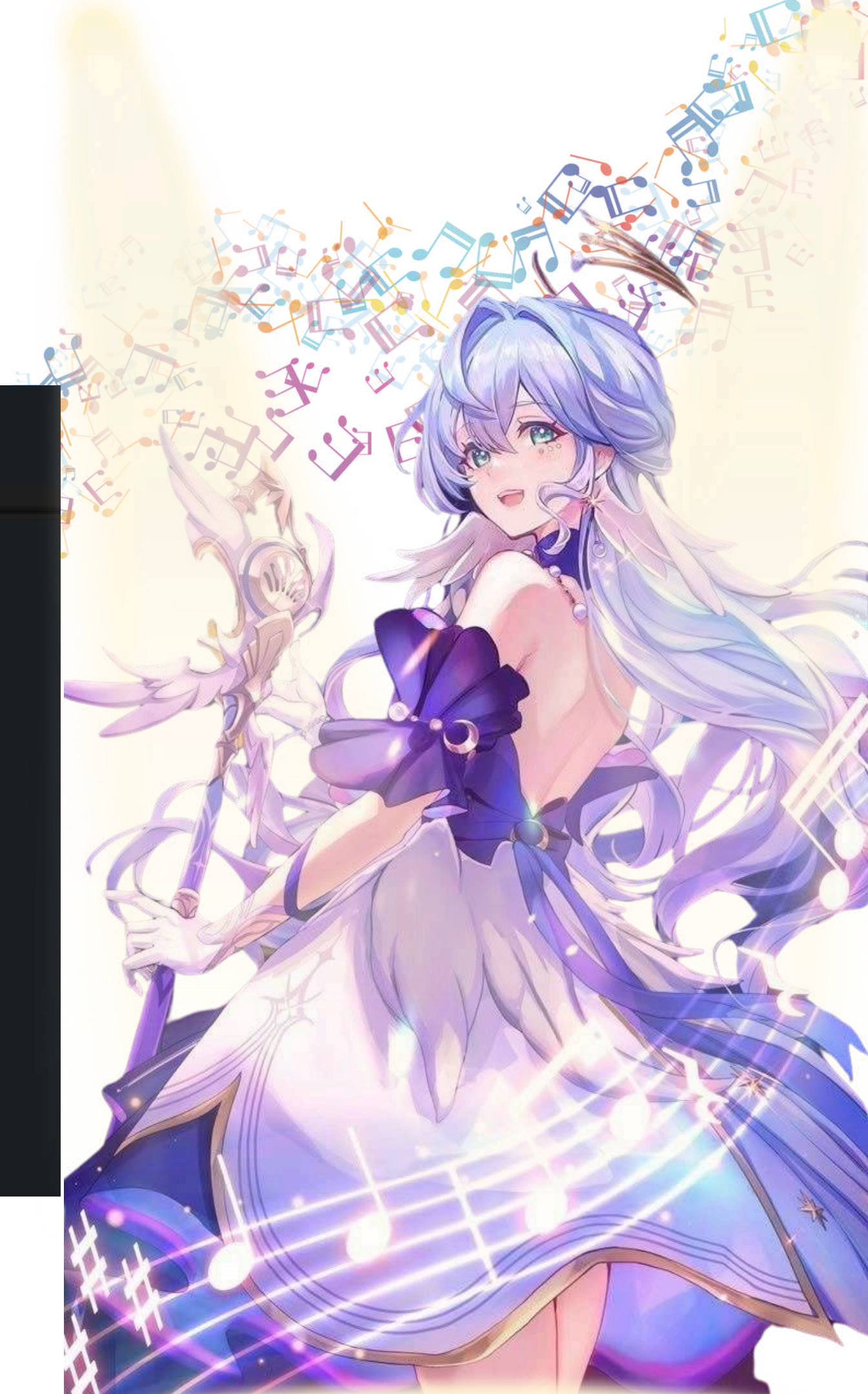


空房搜尋

確保了不會發生重疊訂房的問題

SQL

```
SELECT * FROM Room
WHERE room_id NOT IN (
    SELECT room_id FROM Booking
    WHERE NOT (
        date_end < @user_check_in_date
        OR
        date_start > @user_check_out_date
    )
)
```



編輯 / 取消訂單

訂購房間後，可刪除或修改訂單資訊

SQL

```
INSERT INTO Booking (user_id, room_id, date_start, date_end, pay_type)
VALUES (@user_id, @room_id, @date_start, @date_end, @pay_type);
```

SQL

```
DELETE FROM Booking
WHERE booking_id = @booking_id;
```

SQL

```
UPDATE Booking
SET date_start = @date_start,
    date_end = @date_end,
    pay_type = @pay_type
WHERE booking_id = @booking_id;
```



評論

寫入評論、後來管理平台讀取

SQL

-- 1. 寫入新評論 (Insert Review)

```
INSERT INTO Reviews (user_id, room_id, rating, comment, review_date)
VALUES (@user_id, @room_id, @rating, @comment, @review_date);
```

-- 2. 後台撈取評論 (包含使用者名稱與房型)

```
SELECT
    u.user_name,
    rm.room_description,
    r.rating,
    r.comment
FROM Reviews r
JOIN Users u ON r.user_id = u.user_id      -- 關聯使用者
JOIN Room rm ON r.room_id = rm.room_id    -- 關聯房間
ORDER BY r.review_date DESC;
```



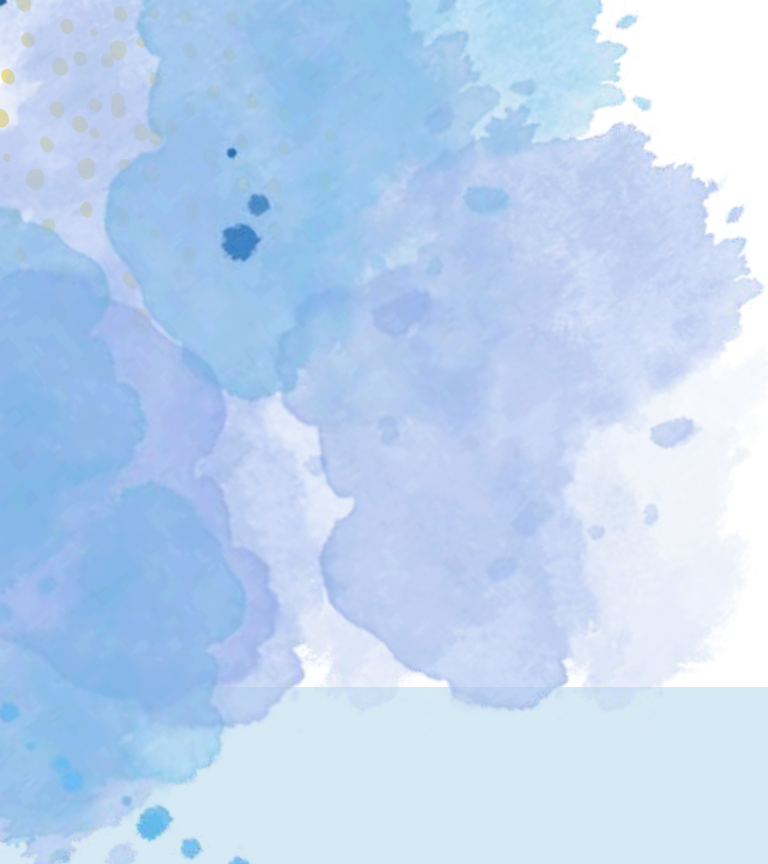
管理員儀表板

統計每月營收

SQL

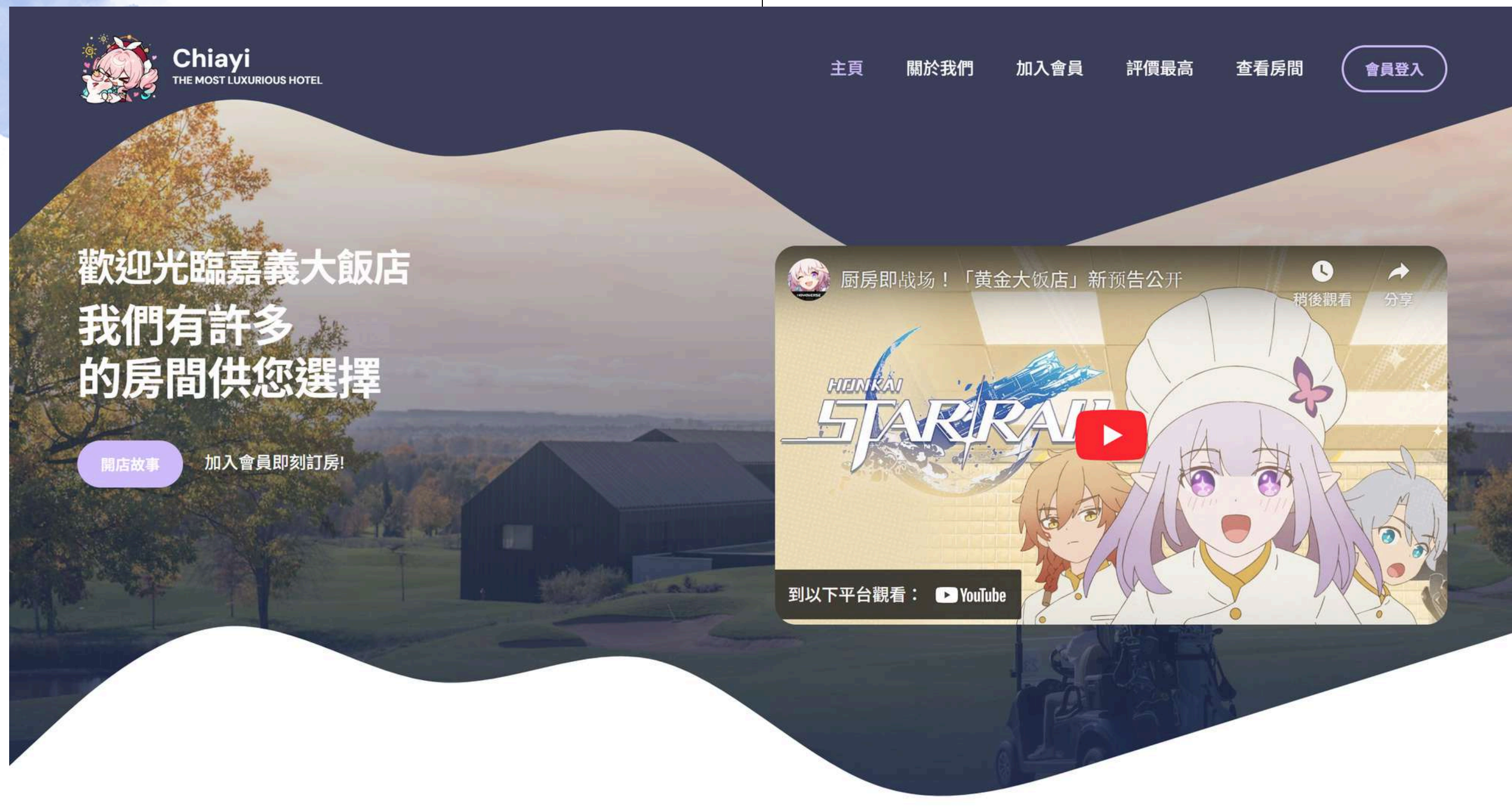
```
SELECT
    FORMAT(b.date_start, 'yyyy-MM') AS month_str,
    COUNT(b.booking_id) AS total_orders,
    -- 計算 (退房-入住天數) * 房價 = 該單總金額
    SUM(DATEDIFF(DAY, b.date_start, b.date_end) * r.room_price)
FROM Booking b
JOIN Room r ON b.room_id = r.room_id
GROUP BY FORMAT(b.date_start, 'yyyy-MM')
ORDER BY month_str DESC;
```





DEMO

點圖片看Demo影片



<https://youtu.be/SJEEjwOD3RQ>